Assignment 8: Monitoring with Prometheus and Grafana

Objective: Implement infrastructure monitoring using Prometheus, Node Exporter, cAdvisor, and Grafana.

GitHub Repository

Repository Name: assignment8-monitoring-prometheus-grafana

Repository Structure

assignment8-monitoring-prometheus-grafana/
■■■ docker-compose.yml
■■■ prometheus/
■ ■■■ prometheus.yml
■ ■■■ alert_rules.yml
■■■ grafana/
■ ■■■ dashboards/
■■■ node-exporter/
■■■ cadvisor/
■■■ screenshots/
■■■ docs/
■ ■■■ architecture.md
■ ■■■ troubleshooting-guide.md
■ ■■■ rollback-guide.md
■■■ README.md

Implementation Steps

1. Installed Prometheus monitoring server.
2. Configured Prometheus scrape targets.
3. Installed and configured Node Exporter for host metrics.
4. Installed and configured cAdvisor for container metrics.
5. Integrated Grafana with Prometheus as a data source.
6. Created monitoring dashboards.
7. Configured alert rules for CPU, Memory, and Service availability.

Monitoring Architecture

Node Exporter ↓ Prometheus ← cAdvisor ↓ Grafana Dashboards ↓ Alert Rules & Notifications

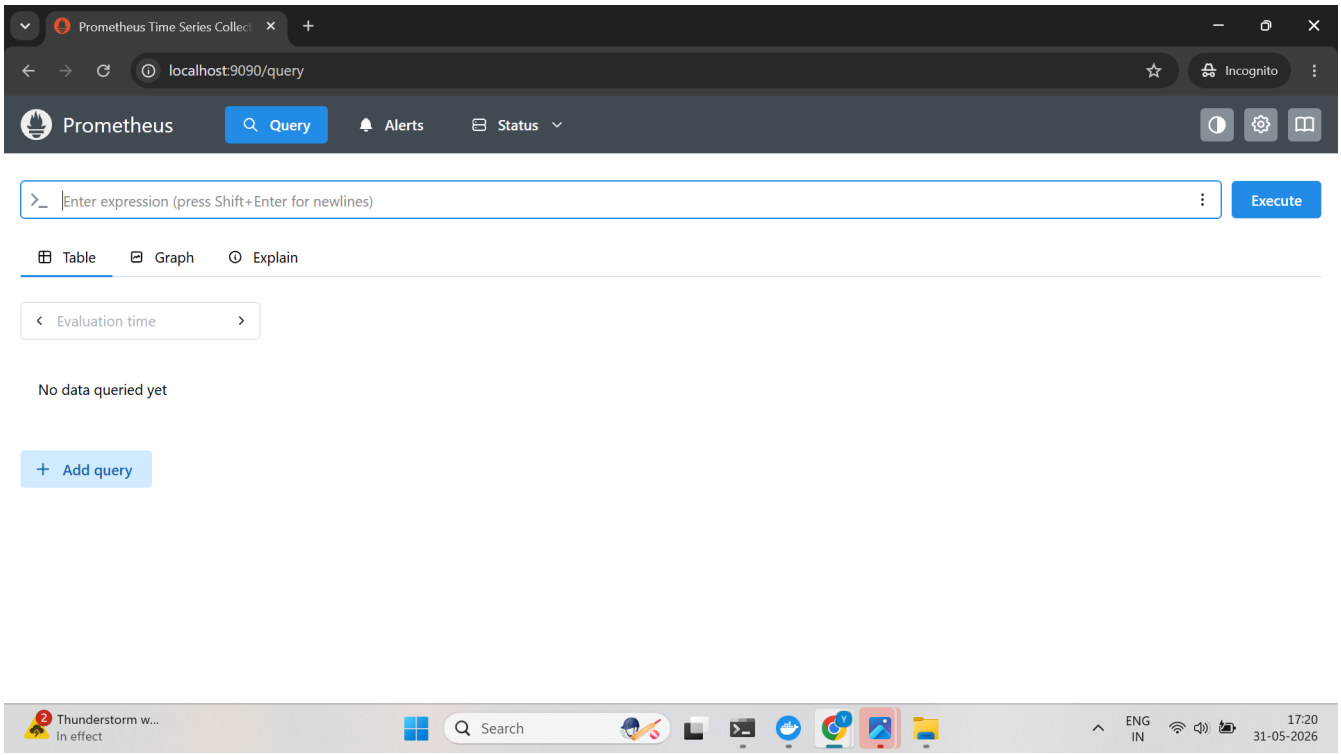
Prometheus Configuration

global: scrape_interval: 15s scrape_configs: - job_name: 'prometheus' static_configs: - targets: ['localhost:9090'] - job_name: 'node-exporter' static_configs: - targets: ['node-exporter:9100']

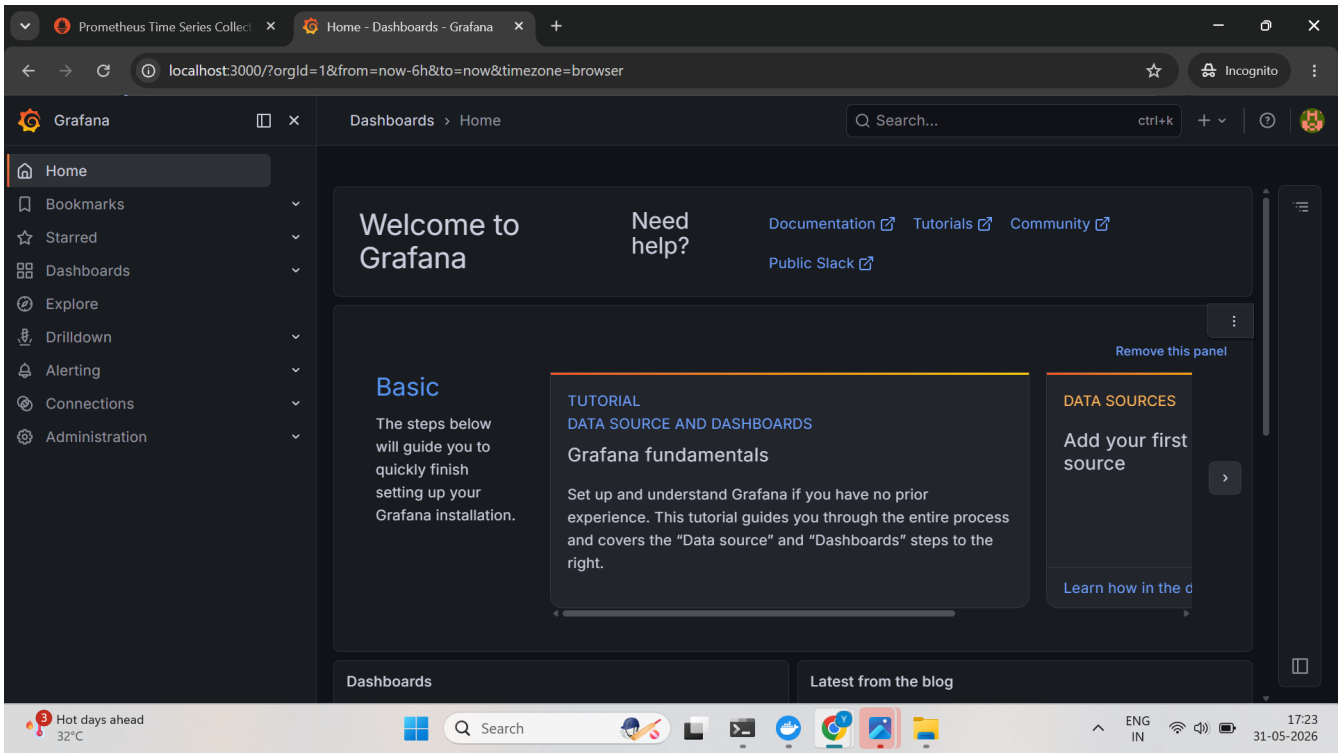
Alert Configuration

groups: - name: monitoring-alerts rules: - alert: HighCPUUsage expr: cpu_usage > 80 for: 1m - alert: HighMemoryUsage expr: memory_usage > 80 for: 1m

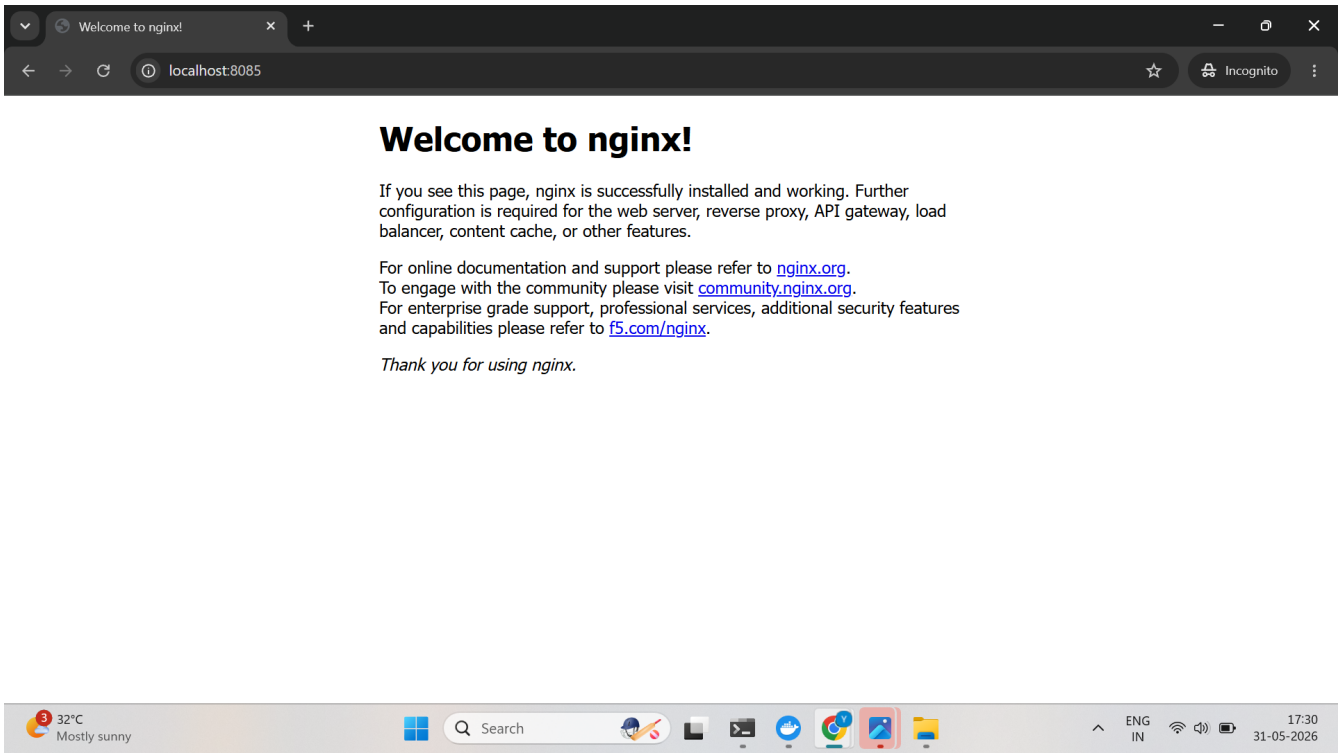
Dashboard / Monitoring Screenshot



Dashboard / Monitoring Screenshot



Dashboard / Monitoring Screenshot



Testing Results

✓ Prometheus server accessible on port 9090.
✓ Grafana dashboard accessible on port 3000.
✓ Monitoring data source configured.
✓ Metrics collection validated.
✓ Alert rules configured successfully.
✓ Monitoring stack operational.

Troubleshooting Guide

Issue: Prometheus target unavailable.
Resolution: Verify exporter service status and scrape configuration.

Issue: Grafana dashboard not displaying metrics.
Resolution: Check Prometheus data source configuration.

Issue: Alerts not firing.
Resolution: Validate alert rule syntax and thresholds.

Rollback Procedure

1. Stop monitoring containers. 2. Restore previous Prometheus configuration. 3. Re-deploy monitoring stack. 4. Verify metrics collection and dashboards.

Key Decisions and Configuration Explanation

Prometheus was selected for metrics collection because of its strong cloud-native ecosystem. Grafana provides flexible dashboard visualization. Node Exporter collects host-level metrics while cAdvisor monitors containers. Alert rules provide proactive monitoring and operational visibility.